1. Resolving Forces

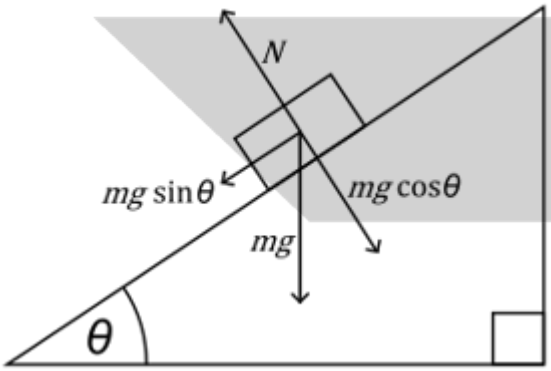
1.1. Resolving Forces

- If force F makes an angle θ with a given direction, the effect of the force in that direction is $F\cos\theta$

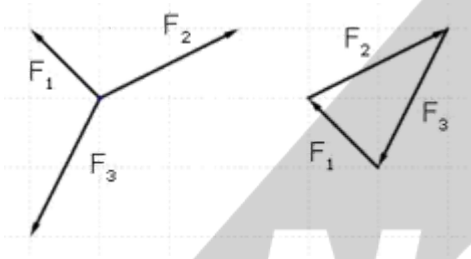
$$F\cos(90-\theta) = F\sin\theta$$

$$F\sin(90-\theta) = F\cos\theta$$

- Forces in equilibrium: resultant = 0
- If drawn, forces will form a closed polygon

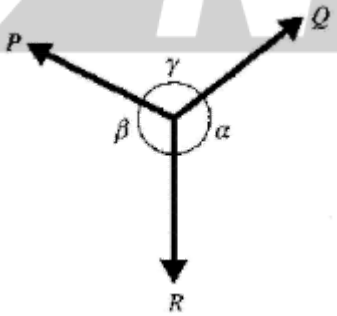


- Methods of working out forces in equilibrium:
  - Construct a triangle and work out the forces
  - Resolve forces in x and y directions; the sum of each = 0



Lami's Theorem:

- For any set of three forces P, Q and R, in equilibrium



$$\frac{P}{\sin\alpha} = \frac{Q}{\sin\beta} = \frac{R}{\sin\gamma}$$

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